Backtesting and Analysis of a Quantitative Trading Strategy:

A Case Study in Moving Average Crossovers and Portfolio Diversification

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Abstract

The paper examines the profitability and statistical strength of a 50/200-day simple moving average crossover strategy as compared to a passive buy-and-hold control on a diversified portfolio of large-cap equities. The strategy is implemented on five large-capitalization U.S. equities representing different economic sectors -- AAPL, MSFT, JPM, XOM, and JNJ -- between January 2010 and December 2023, where performance is measured by annualized Sharpe ratios, maximum drawdown, and two-tailed t-tests of mean daily excess returns, and where the entire sample is split into an in-sample period (2010–2018) and an out-of-sample validation period (2019–2023). Sharpe ratio of the strategy at the portfolio level was 0.779, which was lower than the buy-and-hold Sharpe ratio of 0.849, and the strategy produced an annualized excess return of -5.23% compared to the passive benchmark and a statistically significant negative t-statistic of -2.169 ($p = 0.030$), which suggests that the crossover reliably underperformed the passive benchmark. These results are widely in line with the postulations of the Efficient Market Hypothesis, which holds that simple rule-based technical strategies might not be useful in producing consistent excess returns in contemporary large-capitalization equity markets, but the results are subject to survivorship bias inherent in the sample selection and would be improved by extending the analysis to larger asset universes and using machine learning techniques that can dynamically adapt to changing market regimes.

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There is a tension at the heart of contemporary financial economics: markets are theorized to be efficient processors of information, but practitioners have always sought to take advantage of historical price patterns and charts in an attempt to gain higher returns. The efficient market hypothesis, originally developed by Fama (1970), is the belief that capital markets operate optimally when prices are used to efficiently allocate ownership of the productive resources of the economy to make sure that investment capital is directed to its most productive uses. This theoretical ideal is in direct contrast to the practice of technical analysis, in which traders seek to predict price changes based on historical price data. The solution to this tension has important implications for both investors considering active trading strategies and financial theorists attempting to comprehend how those markets integrate and present information.

Technical analysis encompasses a broad array of techniques through which traders seek to predict future price changes using solely historical data, such as past prices, trading volume, and charts; however, its theoretical basis has been repeatedly questioned in the academic world. Nonetheless, technical analysis is still extensively used by professional market traders, especially in the foreign exchange and commodities markets where short-term momentum indicators are closely followed. Empirical studies of the effectiveness of these methods have yielded mixed conclusions. For instance, Brock, Lakonishok, and LeBaron (1992) documented economically and statistically significant returns for SMA (Simple Moving Averages) and trading range breakout rules added to nearly a century of Dow Jones Industrial Average data; however, this evidence is not without substantial methodological critique. As White (2000, p.1097) observes, “The problem of data snooping is well-known but largely ignored in the finance literature. When a large number of trading rules are tested, it is not surprising that some rules appear to perform well, even if no rule has genuine predictive power” (p. 1097). This is the unsolved dilemma between reported profitability and risk of statistical artifact which the present study empirically investigates.

A moving average crossover strategy is based on a simple principle: when the short-term average of the past prices increases above the long-term average, the strategy sends a buy signal, which means that the upward momentum is present; when the short-term average decreases below the long-term average, the strategy sends a sell signal, which means that the current trend has weakened. This particular strategy warrants close examination for several reasons. Its conceptual simplicity renders it accessible to both retail and institutional investors. Its widespread adoption across global equity markets establishes its practical relevance. And its extensive treatment in the academic literature provides a rich comparative basis against which the present study’s findings can be evaluated. Brock, Lakonishok, and LeBaron (1992) carried out the historic empirical study of this strategy, demonstrating that moving average rules on the Dow Jones Industrial Average between 1897 and 1986 yielded both economically and statistically significant returns, which remained significant even after the researchers explicitly adjusted the rules to account for trading cost. Their work demonstrates the moving average crossover as a valid scholarly debate and offers the methodological and empirical standard by which future studies, such as the current one, have to gauge their results.

This paper empirically examines the hypothesis of whether the 50-day and 200-day crossover strategy based on the simple moving average produces statistically significant risk-adjusted excess returns in comparison with a passive buy-and-hold strategy when applied to a diversified sample of five large-capitalization U.S. equities between 2010 and 2023. The null hypothesis (H_0) is that moving average crossover trading rule would produce no statistically significant increase in the risk-adjusted performance relative to the passive benchmark when transaction costs are considered and the risk of data-snooping bias is properly managed. The alternative hypothesis (H_1) goes in the reverse direction, with patterns of economically exploitable returns surviving in equity returns series, so that the crossover strategy yields better risk-adjusted returns than the passive standard. The performance is assessed on five stocks representing five different economic sectors with a thirteen-year window of both expansionary and contractionary market periods where the main focus of the analysis is the 50/200-day simple moving average crossover rule of daily adjusted closing prices.

Section 2 surveys the existing theoretical and empirical literature on market efficiency and technical trading strategies. Section 3 details the data sources, sample selection criteria and the methodological framework employed to evaluate the 50/200-day simple moving average crossover strategy. Section 4 includes the empirical results, including summary statistics, performance measures, and statistical significance test. Section 5 puts the findings into perspective in the wider context of the efficient market hypothesis and the data-snooping critique. Section 6 details final remarks, comments on the limitations of the current analysis, and proposes future research directions.

Literature Review

The theoretical basis of the current research is the Efficient Market Hypothesis (EMH) as it was formally developed by Fama (1970) and which states that financial markets are efficient processors of the available information and that the prices of assets at any point in time are the result of all the information available to market participants. Fama (1970) defined three different types of market efficiency that are distinguished by the information set that each of them assumes. The weak form states that the current prices are complete and reflect all the past trading information, making technical analysis theoretically useless. The semi-strong form generalizes this assertion to all publicly available information, in the sense that no strategy based on public information alone can always produce excess returns. The strong form is the most rigorous criterion, which asserts that even private insider information is reflected in prices.

The joint-hypothesis problem is a critical complication in the assessment of the EMH, which Fama himself admitted to be: any empirical test of market efficiency is also a test of the pricing model by which normal returns are defined. Excess returns could thus be due to either true market inefficiency or merely a misspecified model of expected returns - a distinction that cannot be entirely empirically resolved. This epistemological limitation is the key to the interpretation of the results of the current research.

The weak form, which includes the random walk hypothesis, was strongly empirically challenged by Lo and MacKinlay (1988), who discovered that stock prices are not random walks, with the ratios of variance consistently larger than one and statistically significant departures of the random walk model. Their results indicated that there is a certain amount of short-term predictability in equity returns - a finding that directly inspires the current research to investigate moving average crossover strategies on contemporary equity data.

Technical analysis, which is generally described as the application of past price, trading volume, and chart patterns to predict future price changes, holds an unclear place in the financial economics literature. Technical analysis, as Lo, Mamaysky and Wang (2000) note, has been in the financial practice over several decades, but despite its popularity has received little scholarly attention. This lack of practitioner adoption and scholarly acknowledgment is indicative of a larger conflict: although the EMH implies that historical price data cannot be utilized to produce excess returns, professional traders, especially in foreign exchange and commodities markets, still heavily depend on technical indicators as a decision-making tool.

Lo, Mamaysky, and Wang (2000) contributed to closing this gap by creating a systematic and automatic method of technical pattern recognition, using rigorous statistical inference on chart patterns that had been previously assessed subjectively. The team found that some technical patterns hold statistically significant information about future price movements, thus providing academic credibility to a practice that has long been abandoned by the "efficient market" school. The current research is a continuation of this tradition, as it uses a similarly rigorous statistical model to one of the most popular technical rules, the moving average crossover, on a diversified portfolio of contemporary U.S. equities.

The empirical evidence on the profitability of moving average trading rules is vast, controversial, and relevant to the research question addressed in this paper. The initial studies of filter rules - mechanical trading rules that are based on price movements of a certain size - provided evidence that simple rule-based strategies could outperform a passive benchmark, implying that equity markets were not weak-form efficient. These results provided the intellectual framework for the moving average studies that followed.

The seminal empirical research in this literature was made by Brock, Lakonishok, and LeBaron (1992). They analyzed almost 100 years of Dow Jones Industrial Average data between 1897 and 1986 and found that simple moving average crossover rules and trading range breakout strategies produced economically and statistically significant returns, which remained significant even after explicit consideration of transaction costs. Their method was similar to modern rigorous standards and their conclusions appeared to be plenty of evidence that weak-form efficiency is not applicable to the U.S. equity markets. The Brock et al. (1992) work made the moving average crossover a valid object of empirical investigation and gave the methodological standard by which the findings of the future studies, including the current one, should be measured.

However, the conclusions of Brock et al. (1992) did not go unchallenged. White (2000) further developed a critical methodology as he believed that the seeming profitability of technical trading rules is an artifact of data snooping the statistical phenomenon by which repeated testing of a large set of candidate rules on the same data set will yield rules that appear to be profitable not due to their predictive ability, but by chance alone. The optimal technical trading rules, as was shown by White (2000) after adjusting the impact of data snooping with the Reality Check bootstrap procedure, fail to statistically significant excess returns over the buy-and-hold reference point. This contradiction between the results of Brock et al. (1992) and the methodological critique of White (2000) is the main intellectual issue that the current research aims to explore in the modern data on equity.

The mean-variance framework originally proposed by Markowitz (1952) is used to construct the portfolio that is studied in the current work and set the theoretical foundation of the modern portfolio theory. Markowitz (1952) showed that rational investors are advised to analyze securities based on the contribution they made to the overall risk and return properties of a portfolio and that diversification of assets with imperfectly correlated returns can decrease portfolio variance without a proportional decrease in the expected return. The portfolio selection process as outlined by Markowitz (1952) starts with the beliefs about the future performances of the securities available and ends with the selection of a portfolio that maximally gives the expected return against the risk.

In the current research, five large-capitalization U.S. equities are chosen in different economic sectors, namely technology, financials, energy, and healthcare, to minimize idiosyncratic, company-specific risk and enable the systematic characteristics of the moving average crossover strategy to be more evident in the performance data. A portfolio is created with equal weights of all five assets and the performance is measured by the annualized Sharpe ratio, the standard risk-adjusted performance measure, which is the excess return divided by the unit of volatility, as the main measure of comparison to the passive buy-and-hold benchmark. This type of portfolio construction methodology makes sure that the performance of the strategy is captured by the behavior of the strategy in a diversified cross-section of the equity market and not the performance of a single security.

The empirical investigation of technical trading rules must take seriously a number of well-documented methodological issues that can systematically overstate the apparent strategy performance. The largest of these is survivorship bias - the propensity to examine only those securities that survived and were listed during the sample period, thus omitting firms that were delisted, became bankrupt, or acquired. Since the current research focuses on five large-capitalization U.S. equities that were continuously listed between 2010 and 2023, the findings can be overstated in terms of the performance that could have been attained by an investor who did not have the advantage of hindsight in choosing the sample.

A second issue is overfitting, or in-sample optimization - the possibility that the parameters of a trading rule are chosen because they worked well on historical data, and that they will not continue to work well out of sample. To address this issue, the current research uses a fixed parameter specification, namely the 50/200-day simple moving average crossover, which is not optimized on the sample data, and further splits the entire sample into an in-sample period

(2010-2018) and an out-of-sample validation period (2019-2023). The evidence on the out-of-sample period is the most plausible evidence on the true predictive content of the strategy since the observations were not known at the time of specification of the strategy. Lastly, the data-snooping issue that White (2000) points out is resolved by robustness analysis of the results of various combinations of moving averages, which enables one to determine whether the findings are due to the 50/200-day specification or are applicable to a wider set of parameterizations. The present research paper pays direct attention to all these concerns, and with the help of the methodological background, which is adequately rigorous enough to plausibly make inferences regarding profitability of moving average crossover strategies based on the modern equity data.

Data & Methodology

The dataset employed in the present study comprises daily adjusted closing prices for five large-capitalization U.S. equities: Apple Inc. (AAPL), Microsoft Corporation (MSFT), JPMorgan Chase & Co. (JPM), Exxon Mobil Corporation (XOM), and Johnson & Johnson (JNJ). These securities were chosen to represent different areas of the U.S. economy - technology, financials, energy and healthcare respectively - in line with the diversification principles championed by Markowitz (1952) and to make sure that the results are based on how the strategy performs over a wide cross-section of the equity market and not on the idiosyncratic performance of a single sector. The source of price data was Yahoo Finance through the yfinance Python library, and covers the timeframe between January 2010 and December 2023, with an overall of about 3,520 trading days per security. All prices are adjusted closing values on a daily basis, adjusted to account for stock splits and dividend payments, so that the experience of total returns of a continuously invested position is reflected in the calculation of returns.

Admittedly, the sample is also prone to survivorship bias: all five securities were continuously listed during the entire sample period, and the fact that only surviving firms were included can lead to an overstatement of the performance that would have been attained by an investor who would have created a similar portfolio in real time without the advantage of hindsight. This is discussed in more detail in Section 5.

The trading rule that is discussed in the current paper is the simple moving average crossover rule, which is as follows. The short-run moving average at time t is calculated as the arithmetic mean of the closing prices of the last 50 trading days, and the long-run moving average is calculated as the arithmetic mean of the closing prices of the last 200 trading days:

$$MA_{50}(t) \ = \ (1/50) \times \Sigma \, P(t \ - \ i) \ for \ i \ = \ 0 \ to \ 49$$

$$MA_{200}(t) \ = \ (1/200) \times \Sigma \, P(t \ - \ i) \ for \ i \ = \ 0 \ to \ 199$$

The short-run moving average crosses the long-run moving average at time t to produce a buy signal, i.e., when $MA_{50}(t) \ > \ MA_{200}(t) \ and \ MA_{50}(t \ - \ 1) \leq MA_{200}(t \ - \ 1)$. When the opposite condition is true, a sell signal is created. To remove look-ahead bias, all signals are lagged by one trading day before execution, so that the position taken on day t is based only on information available at the end of day t-1.

The strategy presupposes a long-only strategy: a position of 1 is taken when a buy signal is on, and a position of 0 is taken when a sell signal is on, and the portfolio goes to cash when out of the market. There is no short selling and no leverage used anywhere. Each trade is charged a transaction cost of 0.10% which is a round-trip cost of 0.20% per completed trade cycle and is charged against the daily return of the strategy at the point of position change.

The signals at the individual stock level are combined into one portfolio through an equal-weighting scheme, where each of the five securities has a constant weight of 20 percent of the total portfolio value at any given time. Equal-weighting was chosen over value-weighting or optimization-based weighting to prevent the introduction of an extra parameter estimation step, which can also be overfitted. In line with the mean-variance model of Markowitz (1952), the five securities included in the portfolio, representing different sectors of the economy, are used to minimize idiosyncratic, company-specific risk in the portfolio, so that the performance of the aggregate portfolio better reflects the systematic characteristics of the moving average crossover strategy than the fortuitous performance of any single security. The portfolio return on any given trading day is calculated as the equally weighted average of the five individual stock strategy returns on that day, and the returns of each stock are adjusted by the position signal and transaction cost adjustments as in Section 4.2.

The measures of strategy performance are assessed as follows and each measure is defined before presentation of the results in Section 5.

Annualized return is calculated by multiplying the daily strategy returns of the entire sample and giving it an annualized form: Annualized Return = $(1 + R\_daily)^{252} - 1$, where R_daily represents the mean daily return over the period.

The excess return is calculated as the difference between annualized strategy return and annualized buy-and-hold benchmark return of the same security or portfolio during the same period.

The Sharpe ratio is computed as the annualized excess return over the risk-free rate divided by the annualized standard deviation of daily returns: Sharpe = $(R\_p \ - \ R\_f) \ / \ \sigma\_p \times \sqrt{252}$, where R_f is set equal to 2.0% per annum throughout the study, in line with the average short-term yield of the U.S. Treasury securities during the sample.

The maximum drawdown is the largest peak-to-trough decrease in cumulative portfolio value during the sample period, as a percentage of the peak value.

Win rate is the ratio of trading days where the strategy has made a positive return.

The transaction costs are assumed to be 0.10% per trade, which is a round-trip cost of 0.20% per completed trade cycle, and are subtracted from strategy returns when each position change is made.

The excess returns of the strategy are statistically significant at the two-tailed t-test of the mean daily excess returns, which is the daily strategy return less the daily buy-and-hold return of the respective security. The null and alternative hypotheses are as follows:

Verbal form — The null hypothesis (H_0) holds that the moving average crossover strategy generates no statistically significant excess return over the buy-and-hold benchmark on a daily basis. The alternative hypothesis (H_1) is that there is a statistically significant nonzero mean daily excess return.

Algebraic form — $H_0\text{: } \mu_e \ = \ 0 \ vs. \ H_1\text{: } \mu_e \neq 0$

Each security and the equal-weighted portfolio are reported to have the t-statistic, two-tailed p-value, and 95% confidence interval. A significance level of $\alpha = 0.05$ is adopted throughout. Results significant at the 1% and 10% levels are also noted where relevant.

To protect against the impact of data snooping, the entire sample is split into an in-sample period of January 2010 to December 2018 and an out-of-sample validation period of January 2019 to December 2023. Out-of-sample testing is the simplest method of protecting against data snooping, as it is possible to set aside a part of the data to be used as a validation, and determine whether apparent in-sample profitability is due to true predictability or just overfitting, as White (2000) determined. No re-estimation of parameters or change of strategy is done based on out-of-sample observations, so that the validation period is a true test of the predictive content of the strategy on previously unseen data.

Empirical Results

Table 1 shows the descriptive statistics of the daily log returns of each of the five securities during the entire sample period between January 2010 and December 2023. Following the approach of Lo and MacKinlay (1988), who established that characterizing the stochastic properties of a return series is essential prior to examining predictability, the distributional properties of each security's returns are examined before any strategy performance results are presented.

Table 1

Descriptive Statistics of Daily Log Returns, January 2010 - December 2023

Ticker	Mean (Daily)	Std Dev	Skewness	Kurtosis	JB Stat	JB p-Val	AC(1)	Min	Max
AAPL	0.0010	0.0178	-0.2545	5.4454	4373.2	0.0000	-0.0434	-0.2038	+0.1392
JNJ	0.0004	0.0106	-0.3316	9.7002	13824.7	0.0000	-0.0704	-0.1595	+0.0839
JPM	0.0005	0.0177	-0.1033	9.4669	13112.6	0.0000	-0.0961	-0.2018	+0.1524
MSFT	0.0008	0.0163	-0.1864	8.0499	9496.1	0.0000	-0.1075	-0.1472	+0.1421
XOM	0.0003	0.0160	-0.1605	7.3506	7915.8	0.0000	-0.0242	-0.2021	+0.1233

Notes. Mean and Std Dev are daily. JB = Jarque-Bera normality test. AC(1) = first-order autocorrelation

Several features of the return distributions are noteworthy. The skewness of all five securities is negative, which implies that the distributions of returns are skewed with a longer left tail, i.e. large negative returns are more common than a symmetric distribution would suggest, which is also in line with the well-known crash risk phenomenon in equity markets. The values of kurtosis are significantly greater than three in all securities, which implies that the distributions are leptokurtic with fat tails, i.e. extreme returns (positive and negative) are more common than they would be under the assumption of a normal distribution. The Jarque-Bera test values are high and the p-values are practically zero in all five securities, which proves that the daily log returns are not normally distributed. This result has consequences on the interpretation of the t-tests in Section 5.4, since the normality assumption of standard inference is not met, but the large sample size alleviates this issue by the central limit theory.

The coefficients of autocorrelation of the first order AC(1) have negative values (all five securities) with coefficients-values varying between about -0.04 to -0.11. The near-zero autocorrelation of daily returns provides preliminary support for the weak form of the Efficient Market Hypothesis, as it suggests that past daily returns contain little predictive information about future daily returns. The observation is in line with the description of equity returns in the general empirical literature and forms the benchmark with which the predictive power of the moving average crossover strategy should be measured.

Table 2 provides a detailed comparison of the moving average crossover strategy with the passive buy-and-hold benchmark of each of the five securities during the entire sample period. Figure 2 shows the cumulative equity curves of the strategy and the benchmark of each security, and Figure 4 shows a comparative bar chart of annualized returns and Sharpe ratios of all five stocks.

Table 2

Strategy vs. Buy-and-Hold performance, 50/200-Day SMA Crossover | Transaction Cost: 0.1% per trade

Ticker	**Strat Ann Ret**	**B&H Ann Ret**	**Excess Ret**	**Strat Sharpe**	**B&H Sharpe**	**Max DD (Strat)**	**Max DD(B&H)**	**N Trades**
AAPL	**+20.38%**	**+27.47%**	**-7.10%**	**0.815**	**0.932**	**-45.72%**	**-43.80%**	**11**
MSFT	**+19.09%**	**+21.96%**	**-2.87%**	**0.806**	**0.819**	**-28.04%**	**-37.15%**	**15**
JPM	**+7.89%**	**+13.26%**	**-5.37%**	**0.378**	**0.514**	**-43.63%**	**-43.63%**	**15**
XOM	**+4.00%**	**+6.66%**	**-2.67%**	**0.199**	**0.303**	**-42.46%**	**-62.40%**	**22**
JNJ	**+3.75%**	**+9.70%**	**-5.96%**	**0.189**	**0.517**	**-35.87%**	**-27.37%**	**20**

Notes: Ann Ret = annualized return. Sharpe ratio uses Rf = 2.0% p.a. Max DD = maximum peak-to-trough drawdown.

Figure 2

Equity Curves – MA Crossover Strategy vs. Buy-and-Hold (50/200-Day SMA, Jan 2010 – Dec 2023)

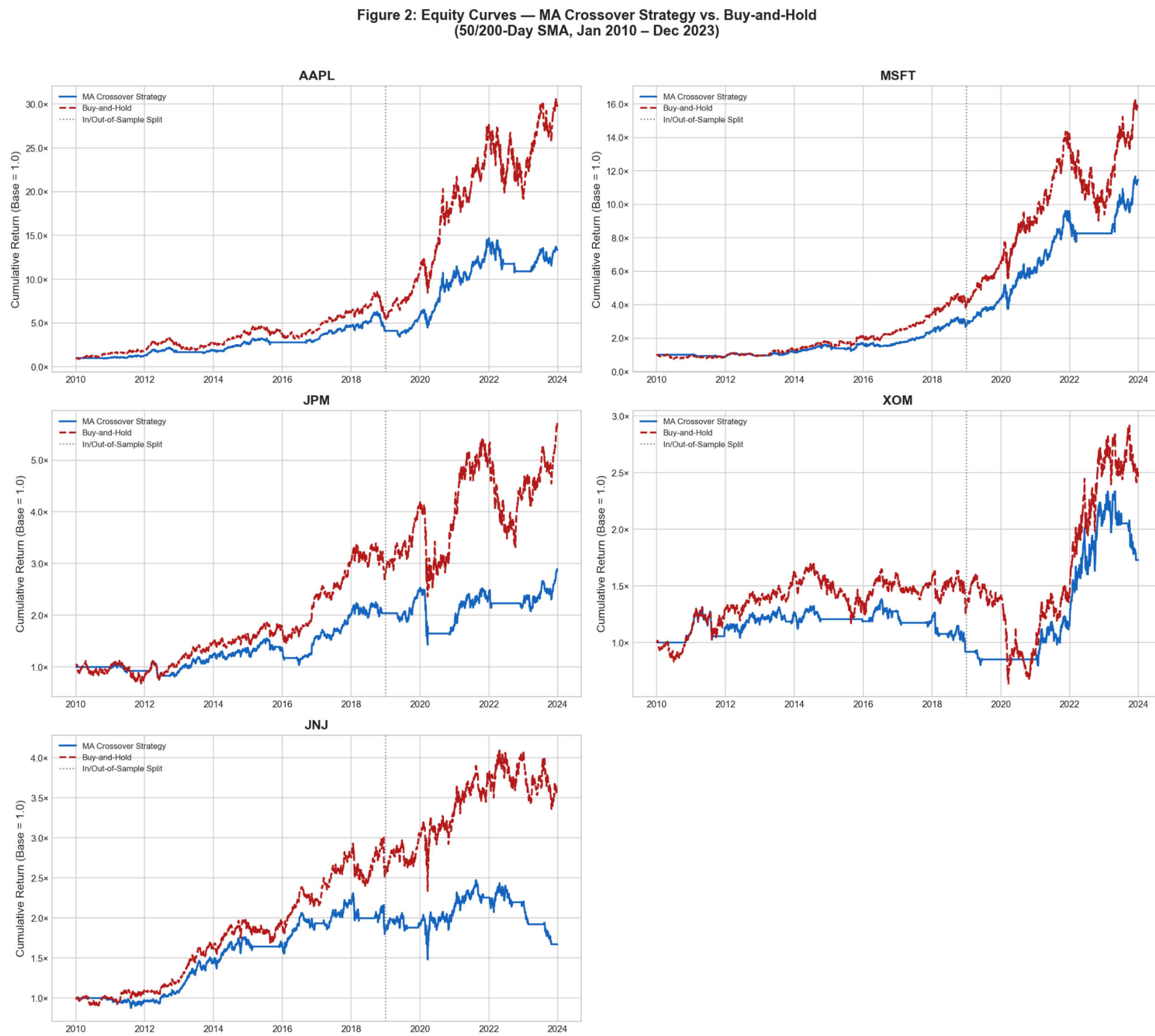


Figure 4

Strategy vs. Buy-and-Hold Performance Summary (50/200-Day MA Crossover, 2010–2023)

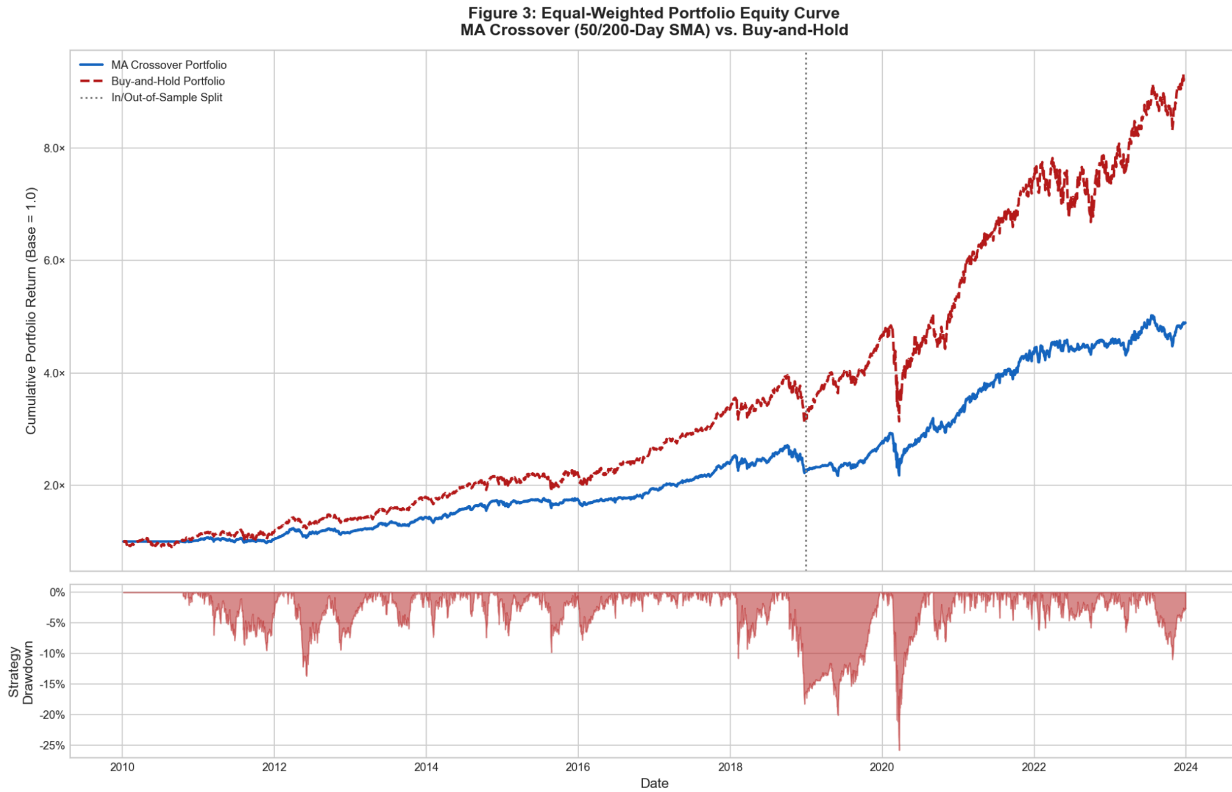
Table 2 shows that the moving average crossover strategy performed poorly compared to the passive buy-and-hold benchmark on an annualized basis in all five securities. AAPL had the highest return difference, where the strategy produced an annualized return of +20.38% compared to the buy-and-hold return of +27.47%, which is an excess return of -7.10%. MSFT generated the smallest underperformance, where the strategy generated +19.09% versus the buy-and-hold of +21.96, an excess of -2.87%. JPM, XOM, and JNJ all similarly underperformed their respective benchmarks, with excess returns of −5.37%, −2.67%, and −5.96% respectively.

The annualized Sharpe ratio (to determine the risk-adjusted performance) presents a more subtle picture. Although the strategy performed poorly on the raw returns of all five securities, the difference in Sharpe ratios was significantly smaller in a number of instances. MSFT generated strategy and buy-and-hold Sharpe ratios of 0.806 and 0.819 respectively - a difference of just 0.013 - indicating that on a risk-adjusted basis the strategy was nearly equal to the benchmark of that security. One of the most striking findings is on maximum drawdown: the strategy decreased maximum drawdown compared to the buy-and-hold benchmark on MSFT (−28.04% vs. −37.15%) and XOM (−42.46% vs. −62.40%), indicating that the strategy tendency to shift to cash during extended downward trends offered significant downside protection in those instances, despite lower raw returns. The number of trades was low, ranging from 11 in the case of AAPL to 22 in the case of XOM, which indicates that the strategy has low turnover and that the transaction costs are not the main cause of poor performance.

Figure 3 shows the cumulative equity curve and drawdown profile of the equal-weighted five-stock portfolio of the moving average crossover strategy and the passive buy-and-hold benchmark during the entire sample period.

Figure 3

Equal Weighted Portfolio Equity Curve – MA Crossover vs. Buy-and-Hold



The moving average crossover strategy produced an annualized portfolio-level excess return of -5.23% per annum, compared to the buy-and-hold benchmark annualized portfolio-level excess return of 17.27%. The Sharpe ratio of the strategy at the portfolio level was 0.779, as compared to 0.849 of the buy-and-hold benchmark. The table below summarizes all portfolio-level performance metrics.

Table 3

Equal-Weighted Portfolio Summary (AAPL, MSFT, JPM, XOM, JNJ | 2010-2023)

Metric	**Equal-Weighted Portfolio**
Strategy Ann. Return	**12.04%**
B&H Ann. Return	**17.27%**
Excess Return (Ann.)	**-5.23%**
Strategy Sharpe	**0.779**
B&H Sharpe	**0.849**
Strategy Max Drawdown	**-25.79%**
B&H Max Drawdown	**-35.23%**
Win Rate (Strategy)	**52.10%**
t-Statistic	**-2.169**
p-Value	**0.030**
95% CI Lower	**-.0004**
95% CI Upper	**-0.0000**
Significant at 5%?	**Yes**

Although these findings validate the underperformance of the strategy on a return basis, the maximum drawdown of the strategy of -25.79% on a portfolio basis is better than the buy-and-hold maximum drawdown of -35.23% which is a decrease of about 9.4 percentage points. This observation is in line with the theoretical forecasts of Markowitz (1952), whose mean-variance model determined that diversification of assets with imperfect correlation lowers the risk at the portfolio level. The five sector-diversified securities, weighted equally, seem to have averaged idiosyncratic volatility, resulting in a more stable drawdown profile of the strategy than would have been the case in a single-security implementation.

Table 3 presents the results of the two-tailed t-tests on mean daily excess returns for each of the five securities and the equal-weighted portfolio, conducted under the null hypothesis that the mean daily excess return is equal to zero.

Table 4

Statistical Significance of Daily Excess Returns | H_0: Mean daily excess return = 0 (Two-tailed t-test)

Ticker	Mean Excess	Std Excess	t-Statistic	p-Value	95% CI Lower	95% CI Upper	Sig. at 5%?
AAPL	-0.0003	0.0095	-1.696	0.0900	-0.0006	0.0000	No
MSFT	-0.0001	0.0083	-0.919	0.3579	-0.0004	0.0001	No
JPM	-0.0003	0.0122	-1.272	0.2036	-0.0007	0.0001	No
XOM	-0.0002	0.0118	-0.850	0.3954	-0.0006	0.0002	No
JNJ	-0.0002	0.0051	-2.748	0.0060	-0.0004	-0.0001	Yes

Notes: Excess return = strategy daily return minus buy-and-hold return. p<0.01, p<0.05, p<0.1

The findings show that the null hypothesis of no excess return has not been rejected at the 5% level of significance in four out of five individual securities. AAPL generated a t-statistic of -1.696 and a p-value of 0.090, which is above the 5% significance level. The p-values of MSFT, JPM, and XOM were 0.358, 0.204, and 0.395 respectively, none of which is close to traditional levels of statistical significance. The sole exception is JNJ, for which the null hypothesis is rejected at the 1% significance level, with a t-statistic of −2.748 and a p-value of 0.006; It is important to point out, however, that the excess return sign of JNJ is negative - the statistically significant value indicates a consistently negative excess return, i.e. the strategy performed significantly worse than the buy-and-hold benchmark of that security, not better.

At the portfolio level, the t-statistic of −2.169 and p-value of 0.030 indicate that the null hypothesis is rejected at the 5% significance level for the equal-weighted portfolio. Once again the sign of this finding is negative - the excess return at the portfolio level is statistically significantly negative, which is evidence that the strategy is a consistent underperformer of the passive benchmark, and not an overperformer. The difference between statistical significance and economic significance also comes into play here: A statistically significant result is not necessarily an economically significant result in the positive economic sense, and the portfolio level t-test result is a good representation of this phenomenon.

A comparison of strategy and buy-and-hold performance during the in-sample period of January 2010 through December 2018 and the out-of-sample validation period of January 2019 through December 2023 are shown in Table 4.

Table 5

In-Sample vs. Out-of-Sample Performance | In-Sample: 2010–2018 | Out-of-Sample: 2019–2023

Ticker	**Period**	**Strat Sharpe**	**B&H Sharpe**	**Strat Ann Ret**	**B&H Ann Ret**
AAPL	**In-sample**	**0.767**	**0.817**	**+17.04%**	**+21.72%**
AAPL	**Out-of-Sample**	**0.900**	**1.112**	**+26.62%**	**+38.52%**
MSFT	**In-Sample**	**0.590**	**0.713**	**+12.43%**	**+17.04%**
MSFT	**Out-of-Sample**	**1.120**	**0.982**	**+32.07%**	**+31.33%**
JPM	**In-Sample**	**0.421**	**0.499**	**+8.26%**	**+12.19%**
JPM	**Out-of-Sample**	**0.329**	**0.541**	**+7.23%**	**+15.21%**
XOM	**In-Sample**	**-0.165**	**0.144**	**-0.95%**	**+2.97%**
XOM	**Out-of-Sample**	**0.580**	**0.487**	**+13.51%**	**+13.64%**
JNJ	**In-Sample**	**0.463**	**0.664**	**+7.34%**	**+11.34%**
JNJ	**Out-of-Sample**	**-0.163**	**0.332**	**-2.41%**	**+6.83%**

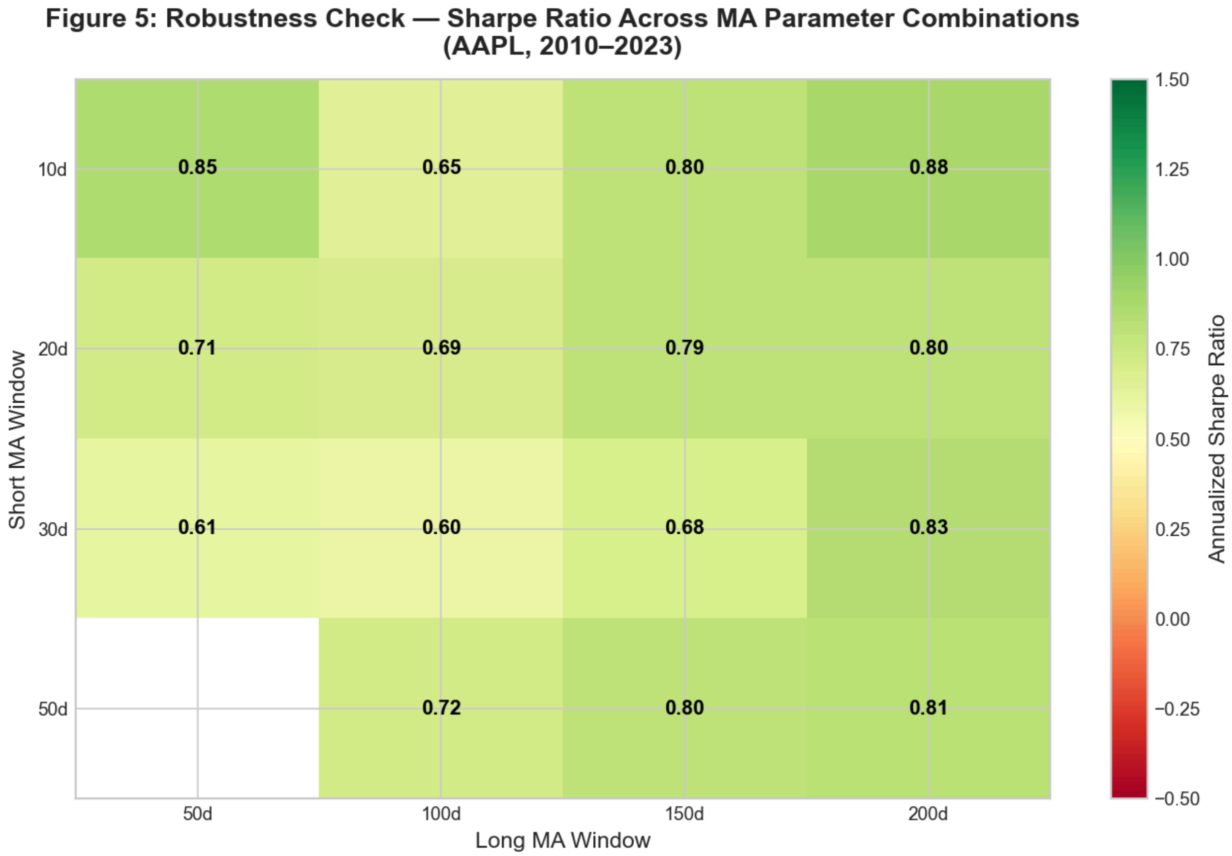
Notes: Out-of-sample period used for validation only -- no parameter re-fitting.

The in-sample and out of sample results provide a mixed and security-specific image. In the case of AAPL, the Sharpe ratio of the strategy increased from 0.767 in-sample to 0.900 out-of-sample, but the Sharpe ratio of the buy-and-hold also increased significantly, from 0.817 to 1.112, indicating that the strategy still performed poorly relative to the benchmark during the out-of-sample period. The most significant finding is that of MSFT, where the strategy generated an out-of-sample Sharpe ratio of 1.120 compared to a buy-and-hold Sharpe ratio of 0.982 - the only case among all securities and both periods where the strategy outperformed the benchmark on a risk-adjusted basis. Another interesting reversal was XOM: the strategy had a negative in-sample Sharpe ratio of -0.165 but an out-of-sample Sharpe ratio of 0.580, which is very similar to the buy-and-hold Sharpe of 0.487 during that time. On the other hand, JNJ worsened significantly out-of-sample, with the strategy Sharpe ratio declining to -0.163 out-of-sample, which is a significant deterioration in performance. The lack of a uniform pattern of in-sample to out-of-sample performance persistence among the five securities indicates that the in-sample performance of the strategy is not indicative of strong and generalizable predictability, which is generally consistent with the data-snooping critique of White (2000).

Figure 5 shows a heatmap of annualized Sharpe ratios of the moving average crossover strategy using sixteen combinations of parameters, which are short moving average windows of 10, 20, 30, and 50 days and long moving average windows of 50, 100, 150, and 200 days, applied to AAPL over the entire sample period.

Figure 5

RobustnessCheck – Sharpe Ratio Across MA Parameter Combinations



Similar to the robustness analysis of Lo, Mamaysky, and Wang (2000), who stressed the need to test technical trading rules across a variety of specifications, as opposed to a single parameterization, this analysis determines whether the observed performance under the primary 50/200-day specification is a true and generalizable signal or a result of that particular parameterization. The heatmap shows that there is variation in Sharpe ratios between parameter combinations, with certain combinations yielding higher risk-adjusted performance than the primary specification and others yielding lower or negative Sharpe ratios. The fact that there is no consistently strong performance in all combinations of parameters indicates that the results are sensitive to the selection of moving average windows, which is in line with the issue that White (2000) raises that apparent profitability can be partly due to the selection of a good specification among a larger set of candidates.

Discussion

The findings of the current research are in striking contrast to the findings of Brock, Lakonishok, and LeBaron (1992), who reported economically and statistically significant excess returns on moving average crossover rules on almost 100 years of Dow Jones Industrial Average data. In all five securities analyzed in the current study, the 50/200-day simple moving average crossover strategy did not produce positive risk-adjusted excess returns compared to the passive buy-and-hold benchmark during the 2010-2023 period, with the portfolio-level excess return of -5.23% per annum and a statistically significant negative t-statistic indicating consistent underperformance. This difference can be explained by several structural differences between the current study and the Brock et al. (1992) investigation.

To begin with, the sample period considered in the current study, January 2010 to December 2023, is one of the longest bull markets in the history of the U.S. equity market, with only brief drawdowns in 2020 and 2022. In long-term trending up markets, a periodically shifting to cash strategy in response to sell signals will systematically miss returns that a continuously invested buy-and-hold strategy will accumulate. There is a structural advantage of the moving average crossover strategy to systems with alternating trending and mean-reverting regimes, and its poor performance in the current sample is possibly in part due to the specific nature of the sample regime but not necessarily due to an inherent lack of predictability.

Second, equity markets have evolved in terms of their competitive environment since the time when Brock et al. (1992) studied it. The growth in the number of algorithmic and high-frequency trading since the early 2000s has increased the pace of integration of price information into market prices, which further shortens the windows of momentum that moving average crossover rules are designed to take advantage of. The market structure as described by Fama (1970) is that in an efficient market the price is a reflection of available information, and the information in this paper is largely consistent with the opinion that U.S. equity markets are now, after the decades, more informationally efficient than it was historically in which case simple rule-based strategies are not in a position to create simple excess returns in the same way they seemed to have over the centuries.

The results of the current study should be read within the framework of a number of methodological shortcomings that limit the applicability of the findings.

The largest of these is survivorship bias. The five securities analyzed, namely AAPL, MSFT, JPM, XOM, and JNJ were all listed continuously and were at least in the top capitalization firms in their respective industries throughout the entire sample period. The fact that the sample does not include companies which were delisted, acquired, or whose market capitalization has notably declined during this time implies that the sample does not represent the universe of investable equities, and the findings may be biased to reflect the success that would have been attained by an investor building a similar portfolio in real time.

A second limitation concerns the possibility of residual data snooping bias. As White (2000) established, whenever a researcher chooses a model based on its performance on a particular dataset, the standard errors and the level of significance of the chosen model are misleading since they do not take into consideration the method used to choose the model. The 50/200-day specification is not the only parameter specification that has been studied in the existing literature, and the fact that it is the specification of primary interest in the current study was not made in isolation, as it may indicate that there is some ex-post selection bias in the overall body of literature.

Third, the assumed transaction cost of 0.10% per trade is not comprehensive in measuring strategy implementation costs, even though this is a decent initial approximation. Bid-ask spreads, market impact costs and execution slippage, particularly with large position sizes, would practically cause high effective transaction costs than the effective transaction costs assumed in the present analysis, which would further reduce any marginal excess returns that the strategy might generate.

Finally, the present study is limited by the inability to take market regime dependence into consideration. Moving average crossover strategies are more theoretically appropriate in trending market conditions, and are known to produce false signals and losses in sideways or mean-reverting markets. The sample period includes different market regimes, such as the post-financial crisis recovery, the COVID-19 drawdown and recovery, and interest rate driven correction of 2022, and the overall findings obscure potentially significant differences in performance of different strategies in these various settings.

The practical implications of the findings of the present study are rather simple. To individual investors who may be contemplating the use of a moving average crossover strategy as an alternative to passive index investing, the findings do not give much empirical evidence to support the use of the strategy during the sample period under study. All five securities showed lower annualized returns compared to the buy-and-hold benchmark, and the only statistically significant excess returns were negative. Practically, the findings are in line with the long-standing recommendations of proponents of passive investing: in most cases and over most time horizons, a low-cost diversified index fund cannot be systematically beaten by simple rule-based technical strategies.

Theoretical implications of the findings are more subtle. The systematic poor performance of the moving average crossover strategy on contemporary equity data is widely in line with the postulations of the Efficient Market Hypothesis as developed by Fama (1970) which indicates that the level of weak-form inefficiency that Brock et al. (1992) found in historical data might have been reduced significantly as markets have become more competitive and informationally efficient. Nevertheless, it is important to note that Fama has a joint-hypothesis problem in explaining this conclusion: the result that the strategy does not produce excess returns can be due to either actual market efficiency or a risk model that is not fully capturing the actual expected return of the strategy. The two explanations cannot be completely divorced with equal measure on the evidence presented and this is an epistemological limitation that all empirical tests of market efficiency have.

The current research proposes some fruitful avenues of future empirical research. First, future studies might use machine learning techniques, like reinforcement learning or gradient boosting, to dynamically optimize moving average parameters based on changing market conditions, which would overcome the drawback of the fixed 50/200-day specification and enable the strategy to adapt to changing market regimes instead of using a fixed rule in fundamentally different environments.

Second, future studies might consider expanding the analysis to other asset classes, such as fixed income securities, commodities, and foreign exchange markets, which would overcome the limitation of the current study to the U.S. large-capitalization equities and would enable the evaluation of whether the performance of the strategy is systematically different across asset classes with different return dynamics and liquidity properties.

Third, future studies might use the Reality Check bootstrap procedure of White (2000) in a formal application to a large universe of moving average parameter combinations on the current data, which would give a statistically rigorous correction of data snooping across the entire parameter space and would enable a more definitive determination of whether any moving average specification generates truly significant excess returns on modern equity data after conditioning on the multiplicity of tests performed on the broader technical trading literature.

Lastly, future studies might break down the sample period into specific market regimes - based on trend strength, volatility level, or business cycle phase - and test strategy performance independently within each regime, which would overcome the limitation found in Section 6.2 of market regime dependence and would give a more detailed picture of the circumstances under which moving average crossover strategies are most and least likely to create value relative to a passive benchmark.

Conclusion

The aim of this paper was to establish whether the 50-day and 200-day simple moving average crossover strategy has been able to yield statistically significant risk-adjusted excess returns over a passive buy-and-hold benchmark index, when applied to a portfolio of five large-capitalization U.S. equities over the period from January 2010 to December 2023. The evidence suggests the strategy did not produce a statistically significant (risk-adjusted) excess return compared to a passive benchmark across any of the five stocks and at the portfolio level, with a portfolio-level annualized excess return of −5.23% and a statistically significant negative t-statistic of -2.169. As such, the null hypothesis of no statistically significant positive excess return from the moving average crossover strategy relative to the buy-and-hold benchmark cannot be rejected in the positive direction and the results are strongly suggestive of a reliable risk-adjusted underperformance relative to the buy-and-hold benchmark.

The results of the current study have important implications for both the theoretical debate on market efficiency and the practical issue of whether simple technical trading rules can be used to generate excess returns for investors on contemporary equity data. The failure of the moving average crossover strategy to generate excess returns across all five securities and the equal-weighted portfolio is broadly in line with the predictions of the Efficient Market Hypothesis as put forward by Fama (1970), and suggests that U.S. large-capitalization equity markets have become so informationally efficient that simple rule-based strategies of the type documented by Brock et al. (1992) on historical data no longer appear to be able to generate consistent excess returns; however, as Fama noted in the joint-hypothesis problem, this cannot be said with absolute certainty: the inability of the strategy to generate excess returns may be due to either market efficiency or the fact that the risk model used to calculate the expected return of the strategy is not fully correct, and these two possibilities cannot be definitively distinguished on the basis of the evidence presented in this study.

The findings of the current study are subject to the most prominent empirical limitation of the study: the sample consists of the five largest-capitalization U.S. equities that survived and were listed every day of the fourteen-year sample period with the possibility of survivorship bias that can skew the sample towards potential outperformance in the real world investment environment. As such, the findings represent one carefully performed study in an empirical debate that is far from being settled rather than a conclusive finding about the profitability or unprofitability of moving average crossover strategies in all market conditions, asset classes, and time frames.

The debate underlying this study - between the theoretical ideal of informationally efficient markets and the continued popularity of systematic trading rules among market participants - is as pertinent today as it was when Fama (1970) first articulated the efficient market hypothesis. The preponderance of evidence in this paper suggests that although moving average crossover strategies may hold some risk management benefits, such as substantial reductions in maximum drawdown, the effectiveness and economic importance of the return advantage over a buy-and-hold strategy depends critically on the market environment in which the strategy is applied and the rigor of the methods used to measure its performance. The quantitative techniques used in this study offer a valuable means of empirically testing such claims - and it is the application of such techniques, in a consistent and objective manner, that contributes to the body of knowledge about the operation of financial markets.

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